



FINAL PROJECT REPORT

Bachelor of Technology in Computer Science and Engineering
Class BTech CSE 25-29

PROJECT TITLE

Village Farm Advisory Kiosk

Assisted Digital Access to Weather, Market and Crop Information for Every Farmer

Case Study No. 31 | AgriTech – Assisted Digital Service

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1. Executive Summary

The Village Farm Advisory Kiosk is an assisted digital service that gives farmers access to weather, market-price and crop information through a trained local operator. Many farmers who are uncomfortable with smartphone applications still need this information to decide when to sow, spray, harvest and sell. The kiosk removes the technology barrier: the farmer simply asks a question, the operator finds the answer from approved sources, explains it in the local language and hands over a printed slip.

The kiosk charges ₹10 per information request and expects 45 requests per day, against monthly operating costs of ₹9,000. At this level of demand, monthly revenue is ₹13,500 and the kiosk earns a surplus of ₹4,500. The break-even point is 900 requests per month, or 30 requests per day.

The model focuses not only on selling information but on building trust. A local Krishi Sahayak operator, sourced and time-stamped answers, expert-reviewed crop advice and a transparent grievance process form the trust model. The pilot will start in one village, prove the service and then expand to nearby villages based on usage, farmer feedback and break-even performance.

2. Problem Statement and Opportunity

Problem Statement

Farmers who are uncomfortable with smartphone applications often still need timely weather, market-price and crop information. Without a simple way to get it, they depend on middlemen, neighbours or guesswork, which leads to missed weather windows, weaker bargaining at the mandi and avoidable crop loss.

Farmers need a simple and reliable system for:

- Knowing the weather forecast before sowing, spraying or harvesting
- Checking current mandi prices before travelling to sell
- Getting practical advice on crop problems and pests
- Understanding government scheme dates and eligibility
- Receiving information in their own language, without using an app
- Trusting that the information is correct and up to date

Opportunity Statement

A village kiosk can convert public information into a trusted, affordable local service. By combining a trained local operator, approved data sources and a small pay-per-use fee, the kiosk can bring digital information to farmers who are left out of app-based services while remaining financially self-sustaining.

The opportunity is to build a low-cost, community-supported information service that combines assisted digital access, farmer trust and regular usage habits.

3. Target Customer Segment

Primary Customers

The primary customers are:

- Small and marginal farmers aged 40 and above with basic or no smartphone use
- Women farmers and tenant farmers with limited digital exposure
- Vegetable and cash-crop growers who are most sensitive to price and weather
- Farmer groups, SHGs and farmer producer organizations

Secondary Stakeholders

Other stakeholders can include:

- Village panchayat and community leaders
- Krishi Vigyan Kendras (KVK) and agricultural experts
- Local traders and input suppliers
- NGOs working with farmers
- Government departments providing weather, price and scheme data

Digital-Access Barriers

- Low literacy and English-only interfaces
- Unfamiliarity with app navigation and fear of making mistakes
- Weak mobile network and cost of mobile data
- Shared family phone or keypad phone only
- Low trust in unknown information sources and fear of fraud

Customer Profile

Example Farmer Persona:

Name: Ramesh

Occupation: Small farmer, 52 years, 2 acres of vegetables and cotton

Requirement: Needs tomorrow's rain forecast, today's tomato price and a remedy for leaf curl, explained in Hindi/Marathi

Pain Points:

- Keypad phone only; cannot use apps
- Travels 12 km to the mandi without knowing current prices
- Misses spraying and sowing windows because of unexpected rain
- Uncertain about advice received through informal sources
- Prefers dealing with someone he knows and trusts

Needs: Simple, affordable, local-language and trustworthy information with human help.

4. Value Proposition Canvas

Customer Jobs

Farmers need to:

1. Decide when to sow, spray and harvest.
2. Choose when and where to sell produce.
3. Identify and solve crop and pest problems.
4. Check scheme and insurance deadlines.
5. Reduce losses and wasted trips.
6. Plan the season with confidence.

Customer Pains

- Cannot use smartphone apps
- Unexpected weather and missed farming windows
- No knowledge of current mandi prices before travelling
- Dependence on middlemen for information
- Fear of wrong or unverified advice
- Mobile data cost and weak network

Customer Gains

- Better selling price and fewer wasted trips
- Timely farming decisions
- Confidence through verified information
- Information in the local language
- Affordable access (₹10 per request)
- A trusted place in the village to ask questions

Products and Services

The Village Farm Advisory Kiosk provides:

- Weather forecast and alerts
- Market-price information
- Crop and pest advice
- Government scheme information
- Operator-assisted voice explanation
- Printed information slip with source and time
- Weekly market-day alerts

Pain Relievers

The model reduces problems through:

- Operator-assisted service for non-app users
- Information in local language
- Approved and verified sources

- Source and timestamp printed on every slip
- Price check before travelling to the mandi
- Complaint register and refund if no answer

Gain Creators

The model creates value by:

- Giving farmers timely, practical information
- Building a trusted local service point
- Keeping the price low at ₹10 per request
- Encouraging regular use through loyalty card and alerts
- Creating a shared digital access point for the village

5. Business Model Canvas

Community-anchored, pay-per-use assisted digital service model.

Key Partners	Key Activities	Value Proposition
• Panchayat, SHGs and FPOs • IMD, Agmarknet, KVK • UPI / payment provider • Internet service provider	• Fetching and verifying information • Assisted request handling • Printing slips and collecting payments • Awareness, alerts and feedback	• Weather, price, crop and scheme information • In local language with human help • Pay-per-use at ₹10 • Trusted, nearby, no smartphone needed
Key Resources	Customer Relationships	Customer Segments
• Trained local operator • Laptop/tablet, printer, internet • Approved data sources • Kiosk space and signboard	• Personal assistance • Loyalty card and weekly alerts • Feedback and complaint register	• Small and marginal farmers • Vegetable and cash-crop growers • Women farmers and SHGs • Farmer groups
Channels	Cost Structure	Revenue Streams
• Village kiosk • Village meetings and SHG networks • Market-day SMS/IVR alerts	• Operator pay, rent, internet, electricity, printing and maintenance = ₹9,000 per month	• ₹10 per information request • Prepaid loyalty cards (cash-flow benefit) • Potential government awareness partnerships in future

6. Kiosk Service Journey

The kiosk should provide a simple end-to-end service that needs no digital skill from the farmer.

Awareness → Greeting & Registration → Request → Information Retrieval → Verification → Voice Explanation → Payment → Printed Slip → Feedback → Record Update

Step 1: Awareness and Arrival

Farmers learn about the kiosk through village meetings, SHGs, wall paintings and panchayat announcements. On arrival, the operator greets the farmer and records name and village in the register (mobile number optional).

Step 2: Request Classification

The farmer states the question in own language. The operator classifies it into one of four information categories:

Category	Content	Source / Refresh
Weather	3–5 day forecast, rainfall, heat and storm alerts for the village block	IMD / state weather service; twice daily
Market prices	Mandi rates for local crops, 2–3 nearby mandis, price trend	Agmarknet / APMC; every morning
Crop information	Sowing time, fertilizer dose, pest and disease remedies, seasonal calendar	KVK, state agriculture department, ICAR advisories
Scheme information	PM-KISAN status, crop insurance dates, subsidy details	Government portals; weekly check

Step 3: Retrieval and Verification

The operator fetches the answer only from approved sources. Crop advice is used only from approved institutions and is reviewed weekly by a KVK or agriculture expert.

Step 4: Explanation, Payment and Slip

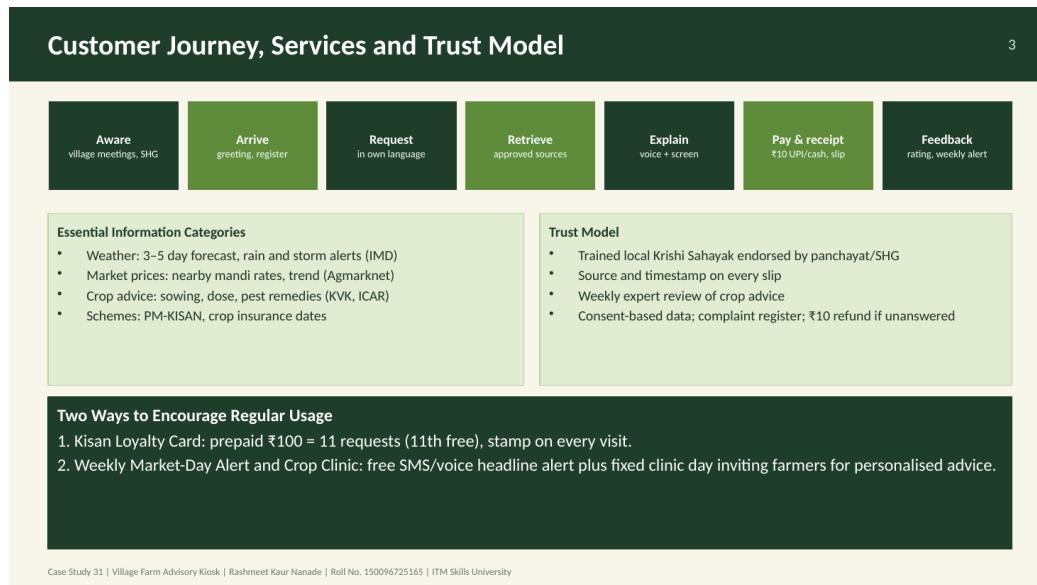
The operator:

1. Explains the answer aloud in the local language.
2. Shows numbers or pictures on the screen.
3. Collects ₹10 by cash or UPI QR.
4. Prints a slip with answer, source, date and time.
5. Records the request in the register.
6. Invites feedback and the next weekly alert.

Trust Model

- Trained local operator (Krishi Sahayak) endorsed by the panchayat or SHG
- Source and timestamp on every answer
- Weekly expert review of crop advice
- Fixed price board and receipt for every payment
- Farmer data never sold and shared only with consent
- Complaint register and ₹10 refund if no answer can be given

Presentation Slide: Customer Journey, Services and Trust Model



7. Supply Chain / Operating Model

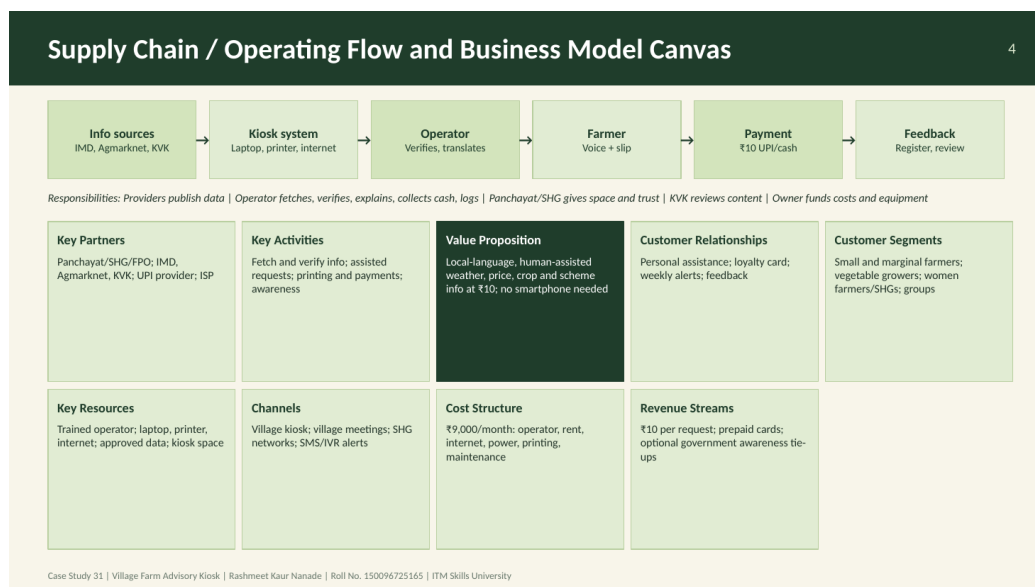
End-to-end flow from information sourcing to delivery to the farmer.

Information Sources → Kiosk System → Operator Verification → Voice Explanation → Payment → Printed Slip → Feedback & Records

Key Responsibilities

Stage / Actor	Responsible Role
Information sourcing	IMD, Agmarknet, KVK and government portals publish the data; kiosk uses only these approved sources
Retrieval and verification	Kiosk operator fetches data, checks source and date, selects the right answer
Explanation and delivery	Kiosk operator explains in local language and prints the slip
Payment and records	Kiosk operator collects ₹10, updates register and daily cash/UPI log
Trust and mobilisation	Panchayat / SHG / FPO provides space, builds trust and brings farmers
Content quality	KVK / agriculture expert reviews crop advice weekly and trains the operator
Equipment and costs	Kiosk owner / coordinator pays costs, maintains equipment and internet, monitors daily reports

Presentation Slide: Supply Chain / Operating Flow and Business Model Canvas



8. Customer Journey and Service Touchpoints

Awareness

Farmers learn about the kiosk through village meetings, SHGs, panchayat announcements and wall paintings.

Arrival and Registration

The operator greets the farmer and records basic details in the register.

Request

The farmer asks a question in own language about weather, price, crop or scheme.

Information Delivery

The operator fetches, verifies and explains the answer aloud with screen support.

Payment

The farmer pays ₹10 by cash or UPI QR and receives a receipt.

Printed Slip

The slip shows the answer, source, date and time for later reference.

Feedback

The farmer rates usefulness with a smiley card and can register a complaint.

Follow-up

The farmer receives a weekly market-day alert and is invited back for the crop clinic.

9. Revenue and Cost Model

The case provides:

- Information price = ₹10 per request
- Expected requests = 45 per day
- Monthly operating cost = ₹9,000
- Assumption: 30 operating days per month; the ₹9,000 is a fixed cost and public data sources have negligible variable cost

Daily and Monthly Revenue

Daily revenue = $45 \times ₹10 = ₹450$

Monthly requests = $45 \times 30 = 1,350$

Monthly revenue = $1,350 \times ₹10 = ₹13,500$

Monthly Surplus

$₹13,500 - ₹9,000 = ₹4,500$

Therefore, the kiosk generates a monthly surplus of ₹4,500, or ₹3.33 per request at 45 requests per day.

Break-even Analysis

Break-even requests = $₹9,000 \div ₹10 = 900$ per month = 30 per day

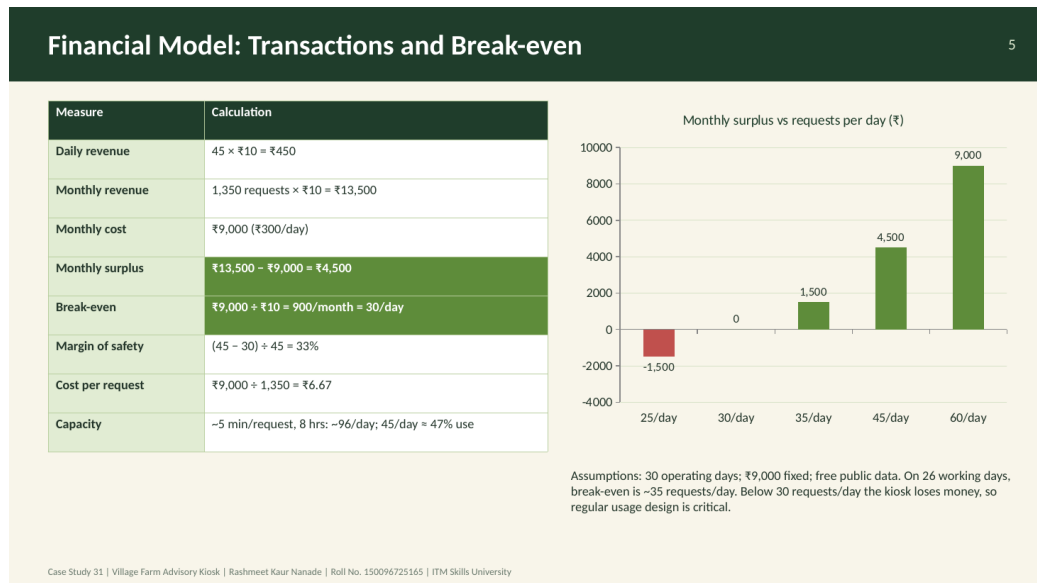
Margin of safety = $(45 - 30) \div 45 = 33\%$

Cost per request at 45/day = $₹9,000 \div 1,350 = ₹6.67$

Important: If the kiosk operates only 26 days a month, break-even rises to about 35 requests per day. Below 30 requests per day the kiosk makes a loss, so regular usage is critical.

Requests per day	Monthly revenue	Monthly surplus / (loss)
25	₹7,500	(₹1,500)
30	₹9,000	₹0 (break-even)
35	₹10,500	₹1,500
45 (expected)	₹13,500	₹4,500
60	₹18,000	₹9,000

Presentation Slide: Financial Model, Break-even and Sensitivity



Capacity Check

At about 5 minutes per request, an 8-hour day allows up to about 96 requests. The expected 45 requests is roughly 47% utilisation, so one trained operator is sufficient for the pilot.

Basic Performance Measures

The kiosk can track:

- Requests per day and break-even achievement
- Revenue per day and monthly surplus
- Capacity utilisation
- Share of requests by category
- Repeat users and loyalty card holders
- Farmer satisfaction rating
- Information error complaints
- Days of service downtime

10. Two Ways to Encourage Regular Usage

1. Kisan Loyalty Card

Farmers can buy a prepaid ₹100 card that gives 11 requests, so the 11th request is effectively free. Each visit earns a stamp. This builds a habit, rewards repeat users and improves the kiosk's cash flow.

2. Weekly Market-Day Alert and Crop Clinic

Registered farmers receive a free weekly SMS or voice alert with a headline price or weather update and an invitation to the kiosk for personalised advice.

Activities can include:

- Weekly market-day price alert

- Fixed crop clinic day in sowing and spraying seasons
- Weather warnings before heavy rain or storms
- Scheme deadline reminders
- Demonstration sessions at village meetings

11. SWOT Analysis

Strengths

- Low price of ₹10 per request
- Trusted local operator
- Meets a real information need
- Low fixed cost
- Works for farmers who cannot use apps

Weaknesses

- Dependence on internet and electricity
- Seasonal variation in demand
- Dependence on operator skill
- Small revenue base
- Manual records may create errors

Opportunities

- Replication in nearby villages
- Addition of scheme and e-services
- Partnerships with KVK and FPOs
- Government digital-literacy initiatives
- Voice-based and regional-language tools

Threats

- Free apps and WhatsApp advice
- Wrong or outdated information
- Low farmer trust or footfall
- Change in data-access policies
- Weather or power disruption

12. Key Risks and Mitigation

Risk	Impact	Mitigation
Wrong or outdated information	Crop loss and loss of trust	Approved sources only; source and time on slip; weekly expert review
Low footfall (below 30/day)	Kiosk runs at a loss	Loyalty card, weekly alerts, SHG mobilisation, market-day focus
Internet or power failure	Service interruption	Mobile hotspot backup, power bank or inverter, cached weather and prices

Risk	Impact	Mitigation
Operator dependence or misconduct	Poor service or cash leakage	Training, backup operator, UPI-first payments, daily cash log
Competition from free apps and WhatsApp	Farmers skip the paid service	Personalised voice help, local-language slips, trust and human support
Data privacy concerns	Farmers stop using the kiosk	Optional registration, consent, no sale of farmer data
Slow adoption by women and older farmers	Lower repeat usage	Women SHG outreach, demonstrations, patient assisted service

13. Recommended Business Model and Go-to-Market Approach

The recommended model is a community-anchored, pay-per-use assisted digital service. The kiosk should start in one village with panchayat or SHG backing and prove the model before expanding.

Phase 1 – Community Formation

Identify farmers and understand their crops, markets, planting seasons, information needs and current sources of information.

Phase 2 – Kiosk Setup

Select and train the local operator, set up equipment, internet, printer and UPI, and finalise approved information sources.

Phase 3 – Pilot Operation

Launch with a village meeting and demonstrations. Start paid requests at ₹10 and introduce the loyalty card.

Phase 4 – Feedback

Collect information on:

- Usefulness and accuracy of information
- Waiting time and operator behaviour
- Most requested categories
- Farmer satisfaction
- Reasons for not returning

Phase 5 – Expansion

Use pilot results to improve services, add weekly alerts and crop clinics, and replicate the kiosk in two to three nearby villages.

14. 90-Day Pilot-to-Scale Plan

Days 1–30: Build

Activities:

- Select village and partner (panchayat / SHG)
- Recruit and train the operator
- Set up kiosk, internet, printer and UPI
- Survey 100+ farmers to map needs
- Finalise information sources and answer templates

Success Criteria:

- Kiosk ready and operator trained
- 100+ farmers surveyed
- 5 information templates ready

Days 31–60: Launch

Activities:

- Soft launch with village meeting and demonstrations
- Start paid requests at ₹10
- Introduce Kisan Loyalty Card
- Log all requests and feedback

Success Criteria:

- 25+ requests per day
- 200 registered farmers
- User rating above 4 out of 5
- 100% of requests recorded

Days 61–90: Pilot and Scale

Activities:

- Start weekly market-day alerts and crop clinic
- Review break-even and monthly surplus
- Fix weak services based on feedback
- Prepare plan for two to three nearby villages

Success Criteria:

- 45 requests per day (minimum 30 to break even)
- 30% repeat users
- Zero major information errors
- Clear expansion plan and demo of the kiosk flow

15. Conclusion

The Village Farm Advisory Kiosk provides a practical assisted digital model that gives farmers affordable, trusted access to weather, market-price and crop information without

requiring them to use smartphone applications. The model combines a local operator, approved information sources, simple payment and regular-usage design.

With ₹10 per request and 45 expected requests per day, the kiosk can generate ₹13,500 in monthly revenue against ₹9,000 of costs, a surplus of ₹4,500, with break-even at 30 requests per day.

The 90-day pilot should focus on proving three things: farmers trust the information, the kiosk reaches at least 30 requests per day, and farmers return regularly. Once these are demonstrated, the kiosk can expand to nearby villages and add more services. A working demo of the kiosk flow accompanies this report.